AdaptiveSats: Improving Institutional Bitcoin Accumulation Strategies

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2026-05-19

1 Executive Summary

Large banks & hedge funds use sophisticated algorithms, privileged market access, and sometimes non-public information to trade. Smaller institutions and retail investors cannot compete on those terms. This project asks: can publicly available Bitcoin (Nakamoto (2008)) on-chain data be used to build a smarter accumulation strategy - one that beats simple Dollar-Cost Averaging (DCA) without requiring any special access or resources?

Bitcoin accumulation strategies are increasingly relevant to institutions seeking long-term exposure to digital assets. However, most accumulation approaches still rely on naive Dollar-Cost Averaging (DCA), where fixed amounts are invested at regular intervals regardless of market conditions. While DCA is simple and robust, it ignores the rich set of publicly available on-chain signals that may provide insight into market valuation, investor behaviour, and network activity.

This project investigates whether publicly available Bitcoin on-chain data can be used to improve accumulation efficiency relative to naive DCA. Using over 41,000 blockchain-derived metrics from the BRK dataset, we aim to design adaptive allocation strategies that dynamically adjust daily investment weights while respecting realistic portfolio constraints.

Our approach combines large-scale exploratory data analysis, feature reduction/selection, clustering, and signal development to identify metrics associated with historically favourable accumulation periods. Candidate strategies include Value Averaging, moving-average-based allocation, and signal-driven adaptive weighting models informed by metrics such as MVRV, NUPL, SOPR, and realized price. Future extensions may incorporate machine learning approaches including Hidden Markov Models (HMMs) among others.

Strategies are evaluated using the Stacksats backtesting framework, where performance is measured primarily through Sats per Dollar (SPD), a metric quantifying Bitcoin accumulated per unit of capital invested. Success is defined by consistently outperforming naive DCA across rolling evaluation windows while passing strict validation checks for leakage, allocation integrity, and robustness across multiple market cycles.

The broader goal of this work is to develop transparent, reproducible, and data-driven Bitcoin accumulation strategies that can be implemented without privileged access or proprietary market infrastructure.

2 Introduction

Traditional financial institutions often rely on sophisticated execution systems, proprietary data pipelines, and privileged market access to optimize investment decisions. In contrast, smaller institutions and retail participants typically operate with limited informational and computational advantages. Bitcoin presents a unique environment where this imbalance is partially reduced: all transaction activity is recorded publicly on-chain, allowing any participant to analyze network behaviour, market structure, and investor activity in real time. Strategies can be audited in real-time on-chain, and the Bitcoin market has operated long enough (since 2009) to provide meaningful historical data across multiple market cycles.

This transparency creates an opportunity to explore whether publicly available blockchain data can improve long-term Bitcoin accumulation strategies. In particular, we investigate whether on-chain signals can identify periods of relative undervaluation and thereby improve capital allocation decisions compared to traditional fixed-schedule investing approaches.

The most common accumulation strategy is Dollar-Cost Averaging (DCA), where investors purchase a fixed dollar amount of Bitcoin at regular intervals regardless of prevailing market conditions. DCA is widely adopted because it is simple, reduces emotional decision-making, and mitigates timing risk. However, it treats all market environments equally and does not account for changing valuation conditions, investor sentiment, or network behaviour. Bitcoin's blockchain provides a large collection of observable metrics that may offer insight into these dynamics. Metrics such as Market Value to Realized Value (MVRV), Net Unrealized Profit/Loss (NUPL), Spent Output Profit Ratio (SOPR), realized price, and various holder behaviour indicators have historically exhibited cyclical relationships with market tops and bottoms. These signals may help identify accumulation periods.

In this project, we use the BRK on-chain dataset, containing over 41,000 metrics spanning market valuation, network activity, holder behaviour, technical indicators, and macro-level blockchain statistics. Because of the dataset's scale and high feature redundancy, we implement a structured feature reduction/selection pipeline involving feature-family groupings, hierarchical clustering, and correlation-based pruning to identify informative signals while reducing dimensionality.

Using this reduced feature space, we aim to design adaptive accumulation strategies that dynamically vary daily allocation weights under realistic investment constraints. Candidate approaches include Value Averaging, moving-average-based trend allocation, and signal-driven models informed by historical on-chain conditions. Performance will be evaluated against naive DCA using rolling-window backtesting and Sats per Dollar (SPD), a measure of Bitcoin accumulated per unit of USD invested.

Ultimately, this project seeks to determine whether transparent, publicly accessible blockchain data can be systematically leveraged to improve institutional Bitcoin accumulation efficiency without requiring proprietary information or high-frequency trading infrastructure.

2.0.1 Objectives

- Identify on-chain/market signals predicting favorable accumulation windows
- Build and validate an adaptive daily allocation model

3 Data & EDA

- **Source:** BRK `brk_metrics.parquet` (Bitcoin Research Kit Contributors (2025) via Hyper-Trial (2025))
- **Schema:** long format with day_utc, metric, value columns
  zero nulls; all values finite

Table 1: Dataset overview

Property	Value
File size	~1.08 GiB
Total rows	236,259,020
Daily observations	6,274 days
Distinct metric keys	41,407
Date range	2009-01-03 to 2026-03-13

The 41,407 metrics span several categories including price, market valuation, holder behaviour, network activity, and technical indicators.

3.0.1 Zero Variance Features Selection

One of the first strategies we used in the feature selection step was removing columns with zero variance. According to Kuhn and Johnson (2019), features with a single unique value (zero variance) provide no information gain. Therefore, removing them helps avoid numerical instability.

After applying this approach, 2,810 features were removed from the training dataset.

	count	mean	std	min	25%	50%	75%	max
10y_dca_days_in_loss	5471.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
10y_lump_sum_days_in_loss	5471.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4y_dca_days_in_loss	5471.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4y_lump_sum_days_in_loss	5471.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5y_dca_days_in_loss	5471.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...	...	...	...
year_2026_unrealized_profit	5471.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
year_2026_utxo_count	5471.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
year_2026_utxo_count_30d_change	5471.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
year_2026_value_created	5471.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
year_2026_value_destroyed	5471.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

2810 rows × 8 columns

Figure 1: Zero-variance features

3.0.2 Identifying Feature Families

Family name	Count of features
UTXO Age Cohorts	18342
Price	6050
Address Distribution by Balance	5250
Age & Halving Cohorts	2737
Entities	2368
Address Activity by Tech Type	1732
Pool-Specific Economics	1612
Network Activity	1415
Holder Cohorts	842
Market & Valuation	370
Benchmarks & DCA	369
Supply & Scarcity	130
Profitability & SOPR	117
Block Reward Distributions	27
Technical Indicators	25
Metadata	22
Grand Total	**41408**

Figure 2: Feature count by family

3.0.3 Halving Events and Price History

Table 2 lists each halving date and the resulting block reward reduction. Because Bitcoin has a hard cap of 21 million coins, the halving reinforces its programmed scarcity. Figure 3 shows how Bitcoin's price (market cap / circulating supply) has evolved across those epochs, illustrating events are generally followed by significant upward price trends. (Subawa et al. (2025))

Table 2: Bitcoin halving events and block rewards

Event	Date	Block reward
1st halving	28 November 2012	25 BTC
2nd halving	9 July 2016	12.5 BTC
3rd halving	11 May 2020	6.25 BTC
4th halving	0 April 2024	3.125 BTC

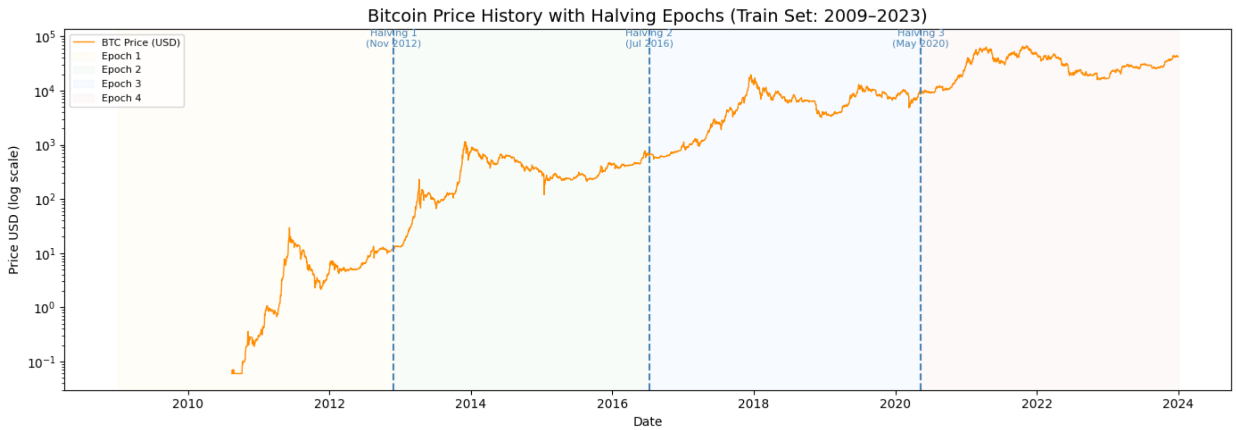

Figure 3: Bitcoin price (market cap / circulating supply) with halving epochs annotated

MVRV, NUPL, and SOPR are key on-chain metrics that have historically signaled favorable accumulation opportunities when they indicate undervaluation (e.g., MVRV < 1, NUPL < 0, SOPR < 1); see Section 7 for full metric definitions. Figure 4 shows how these metrics have varied across different market cycles, with lower values often coinciding with cycle bottoms and prime accumulation zones.

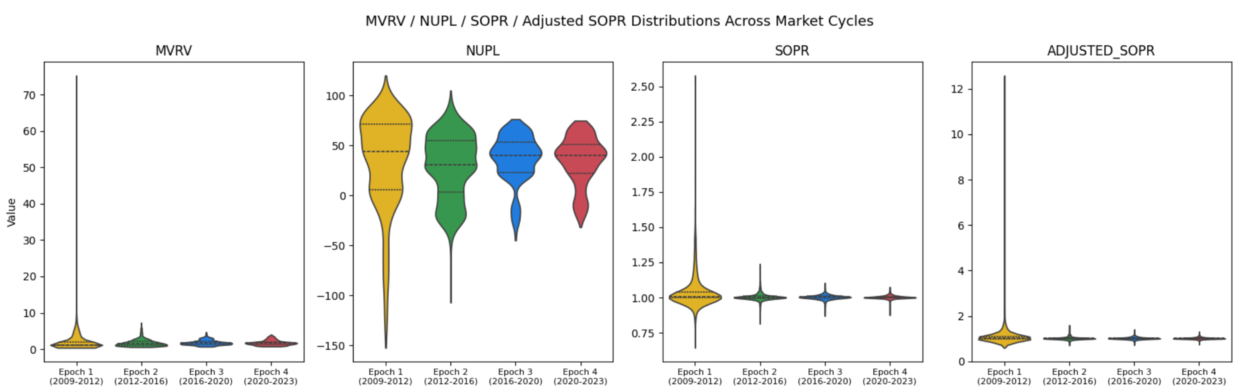

Figure 4: MVRV / NUPL / SOPR distributions across market cycles

- **Limitations:** The dataset requires lazy loading due to its size. Many metrics only begin after 2009, and there is heavy multicollinearity across the 41k features.

3.0.4 Clustering and VIF

After identifying the feature families in our data, we performed hierarchical clustering on each family because we believe there would be large similarities among the members. For each cluster identified, we plan to calculate the VIF, to give us an understanding of the level of multicollinearity present in the data.

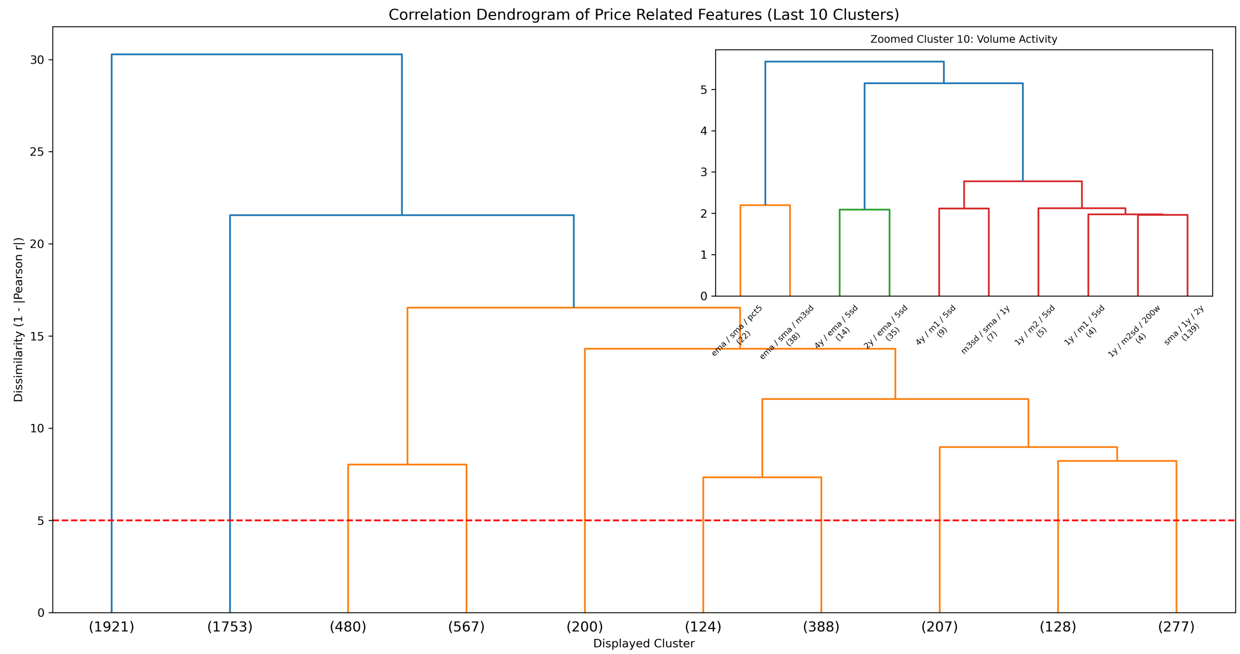

Figure 5: Correlation Dendrogram of Price Features

4 Data Science Approach

4.0.1 Feature Reduction

With over 41,000 features, we aim to identify which features capture similar information to make the dataset more manageable for modeling. To achieve this, we follow a structured feature reduction approach.

First, we group features based on their names using rule based parsing, which helps create feature families such as Price, Benchmark & DCA related features, as depicted in Figure 2. Next, within each family, we apply hierarchical clustering to group features that behave similarly over time, using dendrograms for visualization. Then, from each cluster, we retain a small set of important features and remove those that are highly similar, reducing the dataset size while preserving key information.

This process is computationally intensive when run locally with limited resources, we may explore efficient feature reduction methods in future iterations.

4.0.2 Baseline Strategies

To evaluate our approach, we aim to implement a **Naive Dollar-Cost Averaging (DCA)** strategy as the baseline, where a fixed amount is invested at regular intervals. Our goal is to outperform it by improving accumulation efficiency.

To improve upon this baseline, we are exploring more effective accumulation strategies. In particular, we plan to implement **Value Averaging** as our primary strategy, which adjusts investment based on a target portfolio growth path, allocating more capital during price declines and less during price increases. This aligns well with our objective of maximizing **Sats per Dollar (SPD)**.

In addition, we may also evaluate a **200-day SMA-based strategy**, which adjusts investment based on long-term market trends, providing a simple signal-based benchmark.

4.0.3 Model Development

Our proposed strategy will leverage the reduced feature set to design a data-driven accumulation approach.

The objective is to:

- Identify periods of undervaluation
- Dynamically adjust investment allocation
- Improve accumulation efficiency compared to the defined baseline strategies

4.0.4 Train/Test Split

To avoid look-ahead bias and reflect a realistic backtesting setup, we use a chronological train/test split.

- **Training set:** Data up to December 2023
- **Test set:** January 2024 to March 2026

The test set will remain untouched during model development and will only be used for final evaluation.

4.0.5 Future Work

In future iterations, we may explore additional model such as LSTM (Long Short-Term Memory) based model. This approach will be evaluated against the baseline strategies to assess potential improvements in accumulation efficiency.

5 Evaluation & Success Criteria

5.0.1 Backtesting

The key metric used in backtesting was Sats-per-dollar (SPD). The backtesting process included two frameworks: Validation Checks and Performance Results HyperTrial (2025):

Validation Checks included integrity checks designed to reduce leakage risk and arithmetic drift. These were evaluated through:

1. Weight Sum Validation: Confirms that each window's daily allocations sum to exactly 100% of the budget within floating-point tolerance, ensuring a fair comparison against the naive DCA baseline.
2. Forward-Leakage Test: Future data is masked and the model weights are recomputed to ensure the results remain consistent.
3. Win-rate Requirement: The model must outperform naive DCA in more than 50% of the evaluation windows.

Performance Results summarized the key SPD and percentile results:

1. Percentile Statistics: Measures the minimum, maximum, mean, median, and exponentially weighted percentile (where recent windows receive greater weight) of the winning rate against naive DCA.
2. Strategy Validation: Evaluates whether the model outperforms naive DCA based on the win-rate threshold (greater than 50%), using 2558 rolling one-year windows from 2018 to the present.

The model is then scored by combining the win rate and exponentially weighted percentile into a single benchmark.

5.0.2 Metrics

- **Primary Metric (Sats per Dollar - SPD):**
  We evaluate accumulation efficiency using Sats per Dollar (SPD), defined as the total Bitcoin accumulated per unit of USD invested. A higher SPD indicates better performance, as it reflects a lower average acquisition cost.
- **Smoothness (Allocation Stability):**
  We measure the standard deviation of daily allocation weights. Lower variability indicates a more stable investment schedule with potentially lower market impact.

5.0.3 Success Criteria

A strategy is successful if it achieves higher SPD than the naive DCA baseline in more than 50% of tested rolling windows, is statistically significant ($p < 0.05$), holds across at least one bull and one bear cycle, passes all strict validation checks mentioned above, and has no look-ahead bias.

5.0.4 Stakeholders

- **Trilemma Foundation:** Interested in improving long-term Bitcoin accumulation efficiency
- **Open-source community:** Focused on transparency, reproducibility, and practical usability of the strategy

6 Timeline

Table 3: Project timeline with key milestones

Milestone	Target Date
Train/test split + EDA scripts	May 15, 2026
Feature reduction + baseline DCA	May 22, 2026
ML model + walk-forward CV	May 29, 2026
Backtesting + final evaluation	June 12, 2026
Final report + model	June 23, 2026

7 Appendix: On-Chain Metrics for Long Accumulation Identification

The following table summarises the key on-chain and market metrics and their role in identifying favourable Bitcoin accumulation windows.

Table 4: Key on-chain metrics and their role in long accumulation identification

Metric	Expands to	Definition	Accumulation Signal
mvrv	Market Value to Realized Value	Ratio of market capitalisation to realised capitalisation (the aggregate cost basis of all coins).	Values < 1 (market price below aggregate cost basis) historically mark cycle bottoms and prime accumulation zones.
nupl	Net Unrealized Profit/Loss	NUPL $= \frac{\text{Unrealized Profit}-\text{Unrealized Loss}}{\text{Market Cap}}$; measures the net paper gain/loss of all holders as a fraction of market cap.	Values < 0 indicate the majority of holders are in loss.
sopr	Spent Output Profit Ratio	Ratio of the USD value of coins at the time they are spent (sold) to their USD value when they were created (cost basis). SOPR $= \frac{P_{\text{sell}}}{P_{\text{buy}}}$	SOPR < 1 means coins are being sold at a loss on aggregate, indicating capitulation.
adjusted_sopr	Adjusted SOPR	SOPR calculated after excluding very short-term (< 1 hour) spends to remove noise from exchange transfers and re-organisations.	Same interpretation as SOPR but more reliable.
realized_price	Realized Price	Average USD price at which every coin was last moved on-chain; represents the aggregate cost basis per coin. RP $= \frac{\text{Realized Cap}}{\text{Supply}}$	When market price is below Realized Price, most holders are at an unrealized loss, which has historically signaled favorable accumulation opportunities.

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